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Multi-sensor and MTConnect dataset of metal cutting anomaly in milling from laboratory and industry settings

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This paper presents the Multi-Sensor and MTConnect (MSM) dataset, an open-access resource for anomaly detection in computer numerical control (CNC) metal milling. The dataset integrates synchronized signals from sound sensors, accelerometers, current transformers, and MTConnect-based machine controller data, collected from both laboratory experiments and real industrial production. It covers diverse machining conditions, including normal operations, process anomalies, and tool defects. All data were reviewed and annotated by domain experts using a three-level scheme, enabling consistent labeling across machines and environments. The dataset spans multiple CNC mills, cutting tools, workpiece materials, and cutting conditions, and includes MTConnect information models to ensure semantic consistency and reproducibility. Each dataset unit is time-aligned across sensor modalities, allowing direct use in multimodal analysis. By combining heterogeneous sensors with standardized machine data, the MSM dataset provides a reusable benchmark for artificial intelligence (AI)-based monitoring, anomaly detection, and research in smart manufacturing.

Background & Summary

Advances in the industrial internet of things (IIoT) and artificial intelligence (AI) have created new opportunities for real-time monitoring of machining processes. In computer numerical control (CNC) metal milling, early and accurate detection of anomalies is critical for ensuring product quality and preventing failures of tools, workpieces, and machines[1]. Cutting anomalies often manifest as excessive cutting forces and vibration, and can arise from various causes including chatter[2], forced vibration[3], improper fixturing[4], or tool condition[5], either individually or in combination. The development of robust AI models for anomaly detection depends heavily on the availability of high-quality labeled datasets[6,7].

In response to the growing demand for data-driven monitoring, open-access milling datasets have been released that combine multi-sensor measurements with tool condition and wear levels[8–13]. These datasets provide sensor signals such as cutting forces, vibration, acoustic emission (AE), sound, current consumption, and temperature. They have served as valuable benchmarks for tool wear estimation and condition monitoring. However, most of them have been collected under controlled laboratory conditions with an emphasis on tool wear progression, which limits their ability to capture the variability and complexity of production environments. In contrast, two recent datasets have specifically addressed anomaly detection in milling. The Bosch CNC machining dataset[14] is, to date, the only open benchmark collected directly from an industrial production line. It provides long-term vibration data from three brownfield CNC machines over a two-year period, demonstrating the feasibility of large-scale data acquisition in real manufacturing. However, its focus remains on accelerometer measurements without standardized machine controller information. More recently, the KIT multimodal dataset[15] introduced a combination of force, acceleration, and controller signals for anomaly detection, but it was collected under laboratory conditions on a CNC machine tool. Collectively, these resources illustrate both the progress achieved and the remaining gaps, highlighting the continued need for datasets that integrate diverse

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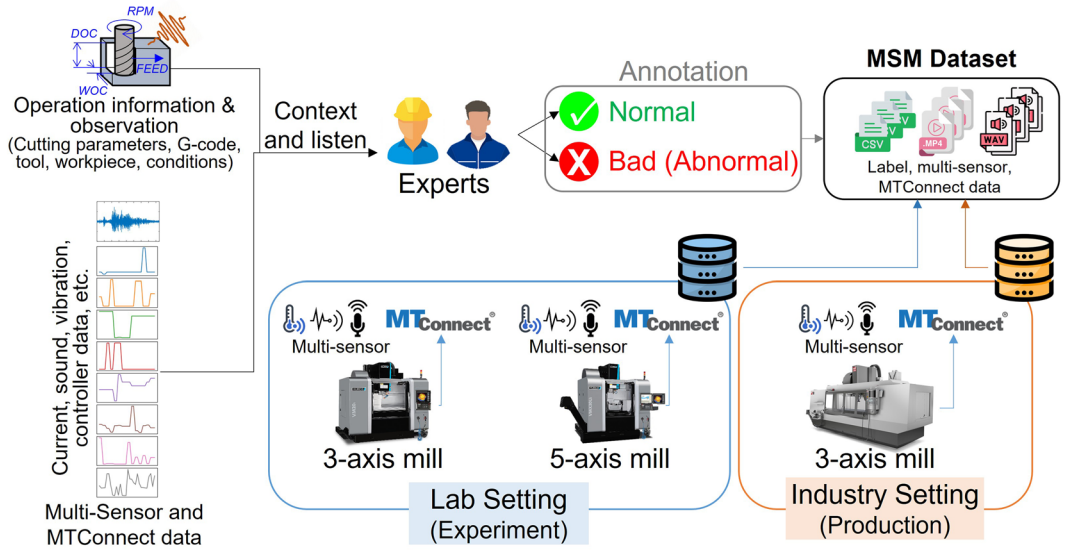


Fig. 1 Overview of the multi-sensor and MTConnect (MSM) dataset.

sensor modalities, standardized machine information, and consistent annotation across both laboratory experiments and industrial production settings.

In real production, particularly in small- and medium-sized enterprises (SMEs), many machines are not equipped with expensive data acquisition systems or sensors. Moreover, profit margins per part are often limited, making it difficult to justify investments in advanced monitoring infrastructure[16]. Even when sensors are available, their type and accessibility often vary across machines, with some providing only limited signals such as spindle current or vibration, while others offer no direct access to internal controller data. Nevertheless, in everyday milling operations, anomalies such as cutting instabilities and misloading issues occur frequently. Operators typically detect these anomalies by stopping the machine, identifying the cause, and taking corrective actions. Yet during production, operators are often occupied with other tasks such as part setup and quality inspection, or may be responsible for simultaneously supervising multiple machines. Such circumstances often make continuous manual monitoring infeasible. As a result, anomalies are sometimes detected only after damage has already occurred, underscoring the importance of automated, real-time anomaly detection for safe and stable operation.

To address these challenges, we developed the Multi-Sensor and MTConnect (MSM) dataset, designed as a comprehensive benchmark for anomaly detection in metal milling. The dataset integrates multimodal data streams, including sound sensors, accelerometers, current transformers (CTs), and MTConnect-based machine controller signals, collected from multiple CNC machines, cutting tools, and workpieces in both laboratory and industrial settings (Fig. 1). MTConnect, an open standard for structured machine data exchange, provides a semantic vocabulary and information model that ensures interoperability across heterogeneous systems. In the MSM dataset, controller-based and low-frequency signals are captured via MTConnect, while high-frequency sensors (sound and accelerometers) are independently synchronized. Unlike datasets tailored primarily to tool wear monitoring, the MSM dataset is designed for general anomaly research in milling. Expert annotations were applied using a three-level labeling scheme: 0 for normal cutting, 1 for anomalous cutting that should be stopped immediately, and 2 for defective tools (e.g., broken insert, flute) intentionally set up for experiments. Importantly, the industrial portion of the dataset was collected from real production operations rather than artificial experiments, ensuring that real-world variability is captured.

The MSM dataset supports a wide range of research directions. It provides a benchmark for anomaly detection in metal milling. It also enables advanced AI approaches such as transfer learning[17], domain adaptation between lab and industry environments[18], cross-modal transfer[19], and multimodal learning[20]. Furthermore, by releasing both sensor data and MTConnect information models, the dataset can open opportunities for large language model (LLM)-based research that leverages structured machine data semantics for context-aware monitoring[21,22]. The MSM dataset offers an open-access benchmark that bridges laboratory and industrial settings, integrates diverse sensor modalities through MTConnect, and provides expert annotations to advance AI-driven monitoring in manufacturing.

Methods

We collected datasets from both laboratory and industrial settings. The laboratory datasets were acquired through controlled cutting experiments at the Indiana Next Generation Manufacturing Competitiveness Center (IN-MaC), Indiana Manufacturing Institute (IMI), Purdue University, using two CNC vertical milling machines. One is a 3-axis CNC mill (VM20i, Hurco), referred to as *imi_vm20i*. Another is a 5-axis CNC mill (VMX30Ui, Hurco), referred to as *imi_vmx30ui*. For the industrial setting, data were collected during real production at the TMF Center (Williamsport, Indiana, USA) from a 3-axis CNC mill (VF-10, Haas), referred to as *tmf_vf10*.

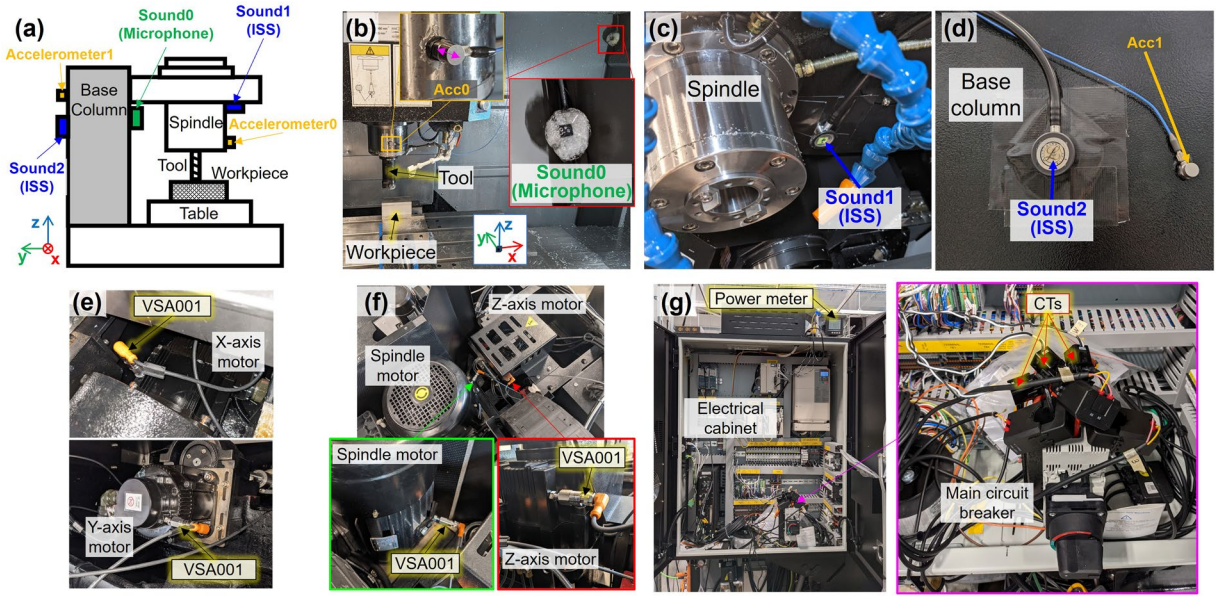


Fig. 2 Sensor deployment on *imi_vm20i*: (**a**) Schematic of the machine and sensor placement, (**b**) Cutting area with microphone (Sound0), and accelerometer on the spindle (Acc0), (**c**) ISS (Sound1) near the spindle, (**d**) ISS (Sound2) and accelerometer (Acc1) on the base column, (**e**) Vibration sensors on X-axis motor (top) and Y-axis motor (bottom), (**f**) Vibration sensors on spindle motor (bottom left) and Z-axis motor (bottom right), and (**g**) Power meter and CTs on main power line (right).

Cutting anomalies in the laboratory datasets were identified through real-time observations and post-collection analysis, while those in the industrial dataset were determined solely through post-collection expert review.

Both laboratory machines (*imi_vm20i* and *imi_vmx30ui*) are equipped with Hurco WinMax® MTConnect, which runs on the machine controller. MTConnect provides real-time access to operational parameters such as spindle speed, override, axis load, tool number, and motion information. For the laboratory machines, MTConnect was used to collect machine-controller data and to integrate low-frequency sensor data, including vibration and power meter measurements. In the laboratory datasets, MTConnect was used both to collect machine-controller data and to integrate low-frequency sensor data, including vibration and power meter measurements. The detailed MTConnect data acquisition procedures differ between the laboratory and industrial datasets and are described in the following section. This integration ensures time synchronization between controller and sensor data. For high-frequency measurements, sound sensors were deployed on all machines, and accelerometers were additionally used on *imi_vm20i*.

A summary of the data acquisition characteristics for each machine is as follows:

- *imi_vm20i*: Largest number of sensors and datasets; MTConnect integration (machine controller, vibration, and power meter); three sound sensors and two accelerometers
- *imi_vmx30ui*: Second largest sensor setup, similar data collection to *imi_vm20i*; MTConnect data integration (machine controller, vibration and temperature, and power meter); two sound sensors
- *tmf_vf10*: Industrial production setting, MTConnect data (machine controller); current transformer, and two sound sensors

The following sections describe (i) the sensor deployment and data pipeline for each machine, (ii) the experimental procedures and industry production, and (iii) the data curation and labeling process.

Sensor deployment and data pipeline. To illustrate the sensor configurations, Figs. 2–4 present the sensor deployment for each machine. Figure 5 illustrates the data pipelines of *imi_vm20i* (Fig. 5(a)) *and tmf_vf10* (Fig. 5(b)). The data flow symbols in Fig. 5(a) are common to both machines (*imi_vm20i* and *imi_vmx30ui*), which were installed in the same laboratory environment and share nearly identical data pipelines. The main differences are that *imi_vmx30ui* does not include accelerometers and *Sound1* sensor at the spindle, and it uses different models of vibration sensors and a different gateway model, while the remaining configuration is the same. All the computers in the lab environment (Fig. 5(a)) use a local router network for communication, and a common time server to ensure data synchronization across multiple data sources.

Since the sampling rate of MTConnect machine data differs from that of the high-frequency sensors, these data were collected separately. For high-frequency sampling sensors, two types of sound sensors for all the machines were utilized: a USB microphone (K053, Fifine), referred to as *Sound0*, as presented in Figs. 2–4(b), and an internal sound sensor (ISS) consisting of a stethoscope (Littmann III, 3 M) and the same microphone,

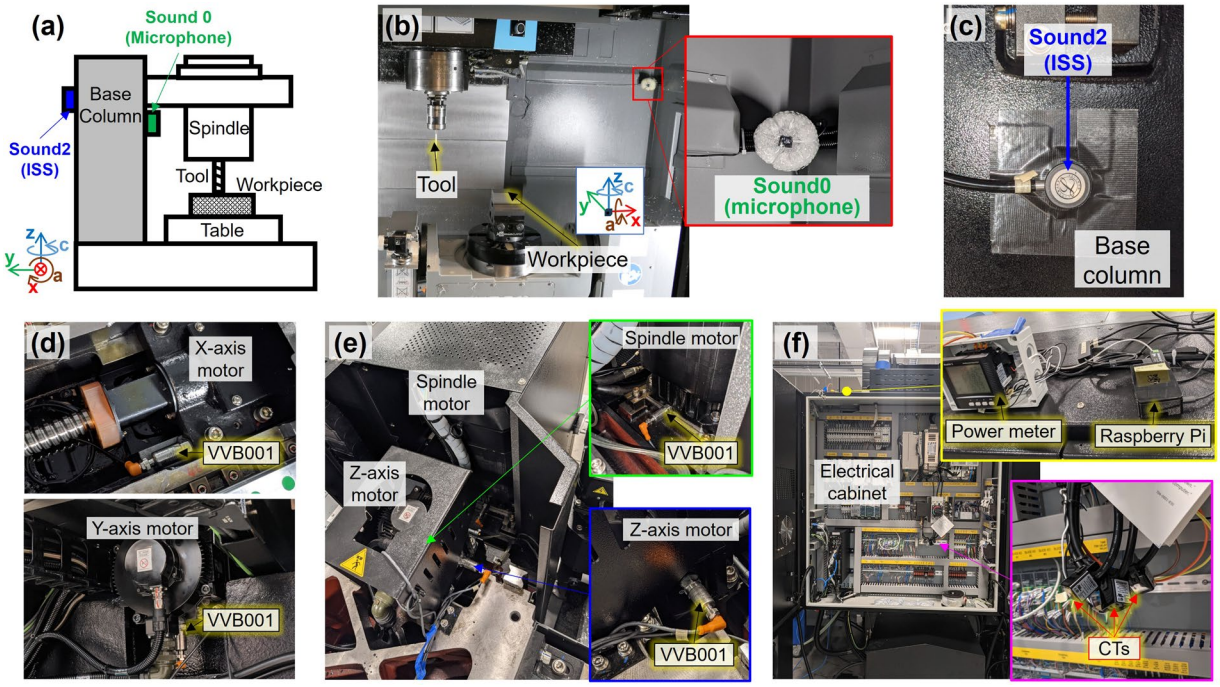


Fig. 3 Sensor deployment on *imi_vmx30ui*: (**a**) Schematic of the machine and sensor placement, (**b**) Cutting area with microphone (Sound0), (**c**) ISS (Sound2) on the base column, (**d**) Vibration and temperature sensors on X-axis motor (top) and Y-axis motor (bottom) (**e**) Vibration and temperature sensors on spindle motor (top right) and Z-axis motor (bottom right), and (**f**) Power meter and Raspberry Pi placed on top of machine (top right), with CTs on main power line (bottom right).

referred to as *Sound1* (only for imi_vm20i in Fig. 2(c)) and *Sound2* (for all the machines presented in Fig. 2(d), Fig. 3(c), and Fig. 4(c)), placed near the spindle and on the base column, respectively. Details of the ISS can be found in[23]. The sound sensors recorded mono-channel audio at a 48,000 Hz sampling rate with 16-bit resolution, and the data acquisition and interaction methods are described in[24]. In the laboratory setting, sound data were stored in WAV format on the local storage of the Raspberry Pi.

On *imi_vm20i* only, two additional high-frequency accelerometers were installed: one on the spindle (*Acc0*, Fig. 2(b)) and another on the base column (Acc1, Fig. 2(d)) as illustrated in Fig. 5(a). Acc0 (352C03, PCB Piezotronics) and Acc1 (352C33, PCB Piezotronics) are single-axis accelerometers connected to a data acquisition system (cDAQ-9178 with NI-9232 module, National Instruments). The DAQ system was connected to a laptop, where data collection was managed by MATLAB script using Data Acquisition Toolbox. The accelerometer data were sampled at 51.2 kHz and recorded in units of mm/s^2 with the timestamp of the first sample. Accelerometer data were stored in CSV format on the laptop as illustrated in Fig. 5(a).

To distinguish them from the high-frequency accelerometers, multiple vibration sensors (VSA001, IFM for *imi_vm20i*; VVB001, IFM for *imi_vmx30ui*) integrated into MTConnect were also employed. These sensors provide time-domain features (e.g., acceleration root-mean-square (RMS), acceleration peak, velocity RMS) as measurement values. To collect data from these sensors, two different gateways were used: VSE953 (IFM, supporting Modbus TCP) for *imi_vm20i*, and AL1322 (IFM, supporting IO-Link) for *imi_vmx30ui*, as indicated in the Sensor column of Table 1. Each gateway was connected to a Windows panel computer on each machine (Fig. 5(a)), which ran an MTConnect adapter to integrate the vibration sensor data into the MTConnect agent on the machine controller. In addition, both laboratory machines were equipped with three-phase power configuration. A power meter (AcuVim CL, AccuEnergy) and three CTs (AcuCT-H040 20:333, AccuEnergy) were installed on main power line. The power meter supports the Modbus RTU protocol and was connected to the Raspberry Pi of each machine. True power consumption (W) and accumulated energy use (kWh) were integrated into MTConnect by running an MTConnect adapter for the power meter on the Raspberry Pi. These sensor data items are summarized in Table 1. The MTConnect adapters were configured to update sensor observations at approximately 1 s intervals. Although the MTConnect agent was queried at a shorter polling interval (approximately 0.1 s) in the laboratory setup, the sensor values themselves were updated according to the adapter update interval.

Note that sensors integrated with MTConnect can provide multiple measurement values. In an MTConnect agent information model (Device.xml), each *Data Item ID* must be unique[25]. For the power meter, the measurement values (true power consumption and accumulated energy use) are uniquely available on both *imi_vm20i* and *imi_vmx30ui*. Accordingly, their *Data Item IDs* are defined as *watt* and *energy* in the information model. For vibration sensors, however, multiple instances of the same measurement type exist on different machine components (e.g., spindle and axis motors). Therefore, each must be assigned a unique *Data Item ID* within the information model. The rule applied in this dataset is to generate the ID by combining the placement abbreviation

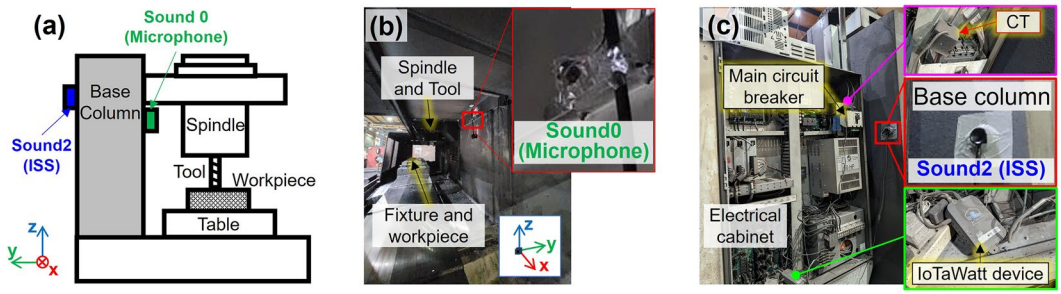


Fig. 4 Sensor deployment on *tmf_vf10*: (**a**) Schematic of the machine and sensor placement, (**b**) Cutting area with microphone (Sound0), and (**c**) CT on main power line (top right), ISS (Sound2) on the base column (middle right), and IoTaWatt device (bottom right).

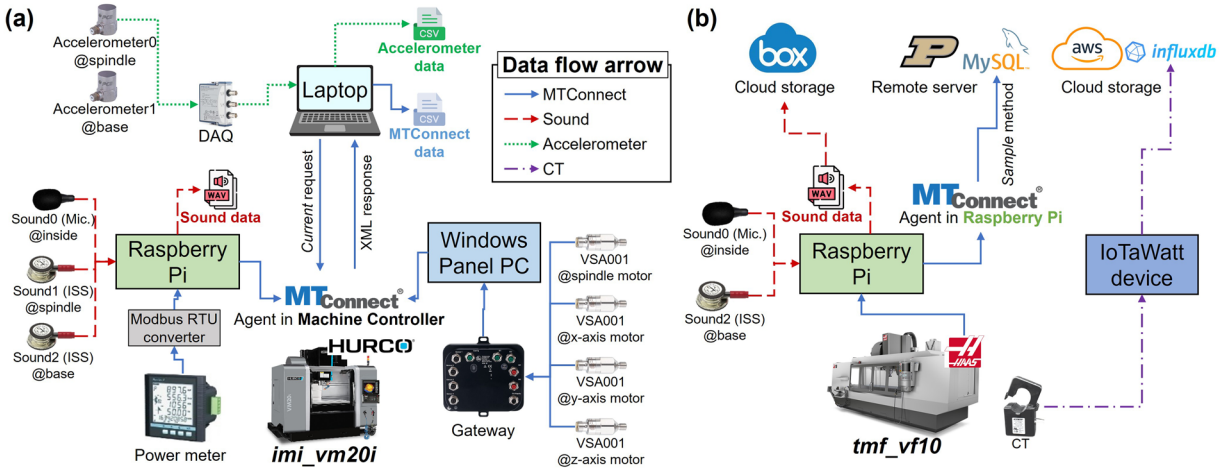


Fig. 5 Data acquisition architecture for (**a**) *imi_vm20i* and (**b**) *tmf_vf10*, showing sensor connections, MTConnect agents, and data flow to storage.

in parentheses (Table 1, Placement column) as a prefix with the measurement abbreviation (Table 1, Data Item column) as a suffix. For example, the crest factor (*crest*) measured on the spindle motor (*sm*) is assigned *smcrest*, and the acceleration RMS (*arms*) measured on the X-axis motor (*xm*) is assigned *xmarms*. The crest factor (CF) is defined as the ratio of the peak amplitude of a signal to its RMS value:

$$\mathrm{CF} = \frac{\max|x(t)|}{\mathrm{RMS}(x(t))}$$

It is commonly used to characterize impulsive behavior in vibration signals and can increase when transient impacts or abnormal events occur during machining processes.

As shown in the MTConnect data flow in Fig. 5(a), the laptop collected the integrated MTConnect data using the *current request* method[25]. In the laboratory setup, the MTConnect adapter provided by the machine manufacturer updated machine observations at approximately 0.1 s intervals. HTTP requests using the *current request* method were repeatedly sent to the MTConnect agent within a polling loop with a similar interval (approximately 0.1 s). Because this process depends on request timing, network latency, and XML parsing, the resulting time intervals between observations in the stored MTConnect data are not strictly constant. The XML responses were parsed into CSV files together with their timestamps, and this procedure was repeated continuously until halted.

The monitoring system at the TMF Center, shown in Fig. 5(b) (right), uses a CT (AcuCT-H100-200:50 mA, AccuEnergy) and device (WiFi Energy Monitor, IoTaWatt) as shown in Fig. 4(c) for *tmf_vf10*. Unlike the power meter, which measures three-phase power consumption, this CT system measures only the current consumption of the first line of the main power supply. The CT monitoring system operates with a 10-second sampling period (0.1 Hz) and stores the data in InfluxDB on Amazon Web Services (AWS). The measurement unit of the CT data is amperes (A). This data could not be integrated into MTConnect. To collect machine controller data from the target machine (VF10, Haas) at the TMF Center, an MTConnect information model was designed and an MTConnect adapter was developed based on the Machine Data Collection (MDC) of Haas Next Generation Control[26]. The MTConnect adapter was configured to update observations at 1 Hz. The collected machine-controller data were retrieved using the MTConnect *sample request* method[25], which returns buffered

Machine	Sensor	Type	Placement	Measurement	Unit	Data Item
imi_vm20i, imi_vmx30ui	AcuVim CL, AccuEnergy	Power Consumption	Main power source	True power Consumption	W	*watt*
				Accumulated Energy Use	kWh	*energy*
imi_vm20i	VSA001, IFM (VSE953, IFM)	Vibration	Spindle Motor (sm), x-axis Motor (xm), y-axis Motor (ym), z-axis Motor (zm)	Acceleration Crest Factor	—	*crest*
				Acceleration RMS	mm/s^2	*rms*
				Acceleration peak	mm/s^2	*peak*
				Velocity RMS	mm/s	*vrms*
				Displacement RMS	mm	*disp*
imi_vmx30ui	VVB001, IFM (AL1322, IFM)	Vibration and Temperature	Spindle Motor (sm), x-axis Motor (xm), y-axis Motor (ym), z-axis Motor (zm)	Acceleration Crest Factor	—	*crest*
				Acceleration RMS	mm/s^2	*rms*
				Acceleration peak	mm/s^2	*peak*
				Velocity RMS	mm/s	*vrms*
				Temperature	°C	*temp*

Table 1. Sensor types, placements, and measurement items integrated via MTConnect. Note: These measurements are reported as MTConnect observations generated by adapters configured with an approximate 1 s update interval.

observations sequentially based on sequence numbers. Each observation retains the timestamp assigned by the MTConnect agent. The Raspberry Pi as illustrated in Fig. 5(b) was connected to the machine via Ethernet and ran MTConnect services. It transmitted the collected MTConnect data to a MySQL database on the Purdue server. Details of the MySQL schema and data collection process can be found in[27]. Two sound sensors (*Sound0* and *Sound2*) were also connected to the Raspberry Pi. The data acquisition procedure was the same as described above. Sound data were continuously collected in 1-minute segments and stored in WAV format on the local storage of the Raspberry Pi. The files were then periodically uploaded to the BOX cloud storage to free local storage space on the device.

The complete MTConnect information models (*imi_vm20i.xml*, *imi_vmx30ui.xml*, *tmf_vf10.xml*) are provided in the data repository. The MTConnect machine-controller variables included in the dataset comprise operational data items such as spindle speed, override, axis load, tool number, axis position, and machine state. The machine-controller variables available in the dataset depend on the MTConnect information model of each machine and can be identified directly from the corresponding Device.xml files. In these information models, each MTConnect Data Item defines the associated machine component, measurement type, and identifier used in the dataset. In this dataset, the sensors integrated through MTConnect include *Description* element per additional data item from sensors in the information model, which enables users to identify sensor placement, sensor model, and measurement values directly from the model itself. Providing these information models is important for ensuring reproducibility and interoperability of the dataset. The MTConnect agent information model (Device.xml) defines the structure and semantics of the data items. By sharing these files, users can directly map sensor data to their corresponding machine components and measurement values without ambiguity. This also enables consistent interpretation across different machines and experiments, since identical measurement types (e.g., acceleration RMS, power consumption) are standardized by unique *Data Item IDs*. Furthermore, the availability of these models facilitates integration with other MTConnect-compliant systems and supports researchers and practitioners in extending the dataset to additional machines or sensors.

Experimental procedure and industry production. The experimental procedures in the laboratory setting and the real production data from the industry are described in this section. To construct the MSM dataset, the experiments were designed to cover a wide range of tools, materials, and machining conditions. Various milling tools with different diameters and flutes numbers (square end mills and shell mills) were used, together with workpieces commonly employed in mechanical part machining (e.g., AA 6061, AISI 1018, AISI 4140).

Figure 6 shows the tools and their corresponding IDs used in the MSM dataset. Tool IDs marked in blue (A, B, C, and E) were used in both *imi_vm20i* and *imi_vmx30ui*, those in yellow (C1 and F1) were used only in *imi_vm20i*, and those in red (C2 and F2) were used only in *imi_vmx30ui*. The purple IDs (G1 and G2) indicate tools used exclusively in *tmf_vf10*. Note that Tool G2 is the only tool with a lead angle (30°), while all other tools have a square cutting cross section. An asterisk (*) following a Tool ID denotes tools intentionally prepared in a bad condition for experimental purposes, as shown in Fig. 7. For example, Tool B* was set up with one of its three inserts completely broken (Fig. 7(a)), and Tool D* was prepared with severe flute edge breakage and chipping (Fig. 7(b), top). Corresponding good-condition tools (e.g., Fig. 7(b), bottom) were also included in the experiments under the same cutting conditions. Detailed specifications of each tool, along with the associated workpiece materials and machines, are provided in Table 2.

Figure 8 presents a workpiece (AA 6061) and schematics of machining paths. In the laboratory setting, raw materials (AA 6061, AISI 1018, and AISI 4140) were purchased as long bars from a vendor (McMaster-Carr) and cut into uniform sizes using a band saw for experiments. The workpiece dimensions ($w \times d \times h$) as shown in Fig. 8(a) were $106.7 \times 50.8 \times 57.4$ mm for AA 6061, $50.8 \times 33.0 \times 34.5$ mm for AISI 1018, and $55.9 \times 38.1 \times 50.8$ mm for AISI 4140, respectively.

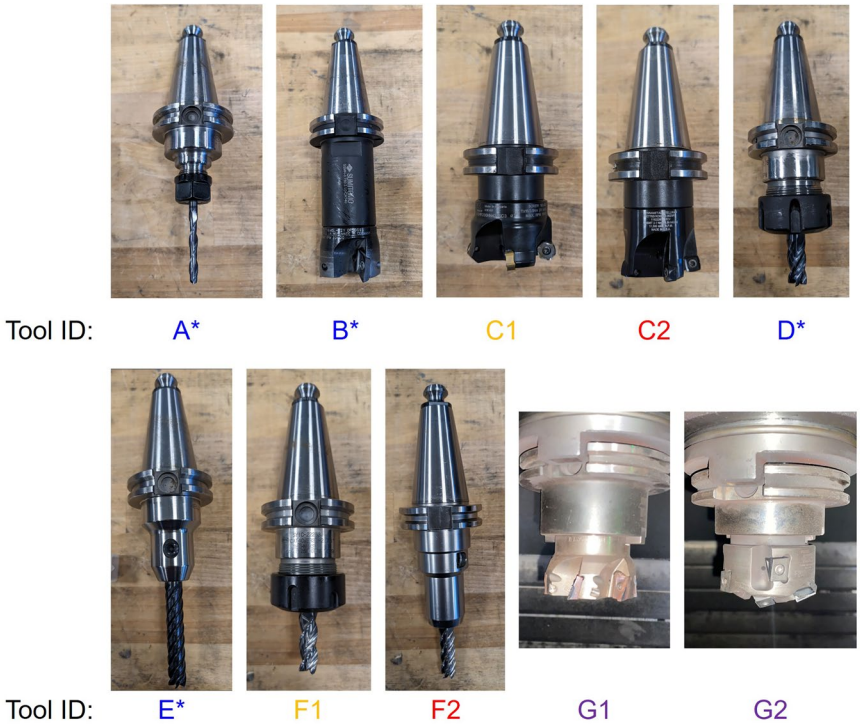


Fig. 6 Photographs of cutting tools with their corresponding IDs used in the MSM dataset. Detailed tool specifications, including tool diameter, flute number, and manufacturer models, are provided in Table 2.

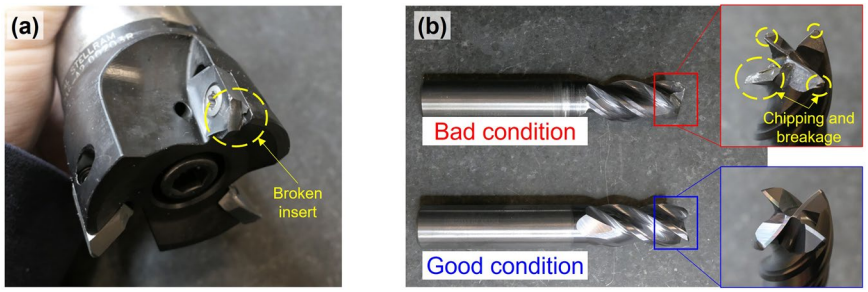


Fig. 7 Experimental setup of bad tool conditions: (**a**) Broken insert of Tool B, and (**b**) Comparison of bad condition (chipping and flute breakage) and good condition (new) of Tool D.

The schematic in Fig. 8(b) illustrates the tool paths and cutting parameters used in the laboratory experiments, including spindle speed, feed rate, cutting direction (up-mill vs. down-mill), depth of cut (a_p) and width of cut (a_e). Before each cutting experiment, a probe system was used to ensure the workpiece and tool origin: an optical tool setter (1/2AA OTS, Renishaw) for tool probing and a modular optical probe (OMP40M, Renishaw) for part probing. As shown in Fig. 8(b), the coordinate origin was set to the negative direction of the x- and y-axes, with margins along the x-axis to include air-cut paths. Note that all laboratory experiments were conducted under wet cutting conditions (coolant flooding), whereas all industrial operations were performed under dry cutting. Cutting parameters (spindle speed, feed rate, a_p, and a_e) were selected based on tool manufacturer's recommendations for the corresponding workpiece materials and adjusted to incorporate diverse machining conditions while considering workpiece size. Since milling instability can be influenced by cutting direction[28,29], a zig-zag tool path was applied to ensure that both up- and down-milling were included under identical conditions. In addition to recommended cutting parameters, some deliberately aggressive cutting parameters were also tested to broaden the range of operating scenarios and to increase the likelihood of inducing anomalies such as chatter or tool failure. During the experiments, even catastrophic cutting failures and severe anomalies were observed. Examples are shown in Fig. 9. Specifically, Fig. 9 (a) illustrates a case of workpiece slippage that caused significant surface damage. The yellow dashed line indicates the initial vertical direction of the workpiece, aligned with the z-axis. Such slippage could be attributed not only to chatter and excessive tool loading but also to fixture failure, which compromised workpiece stability. This instability eventually led to tool breakage, as shown for Tool E in Fig. 9(b). Figure 9(c) shows another anomaly in which residual chips accumulated on the surface of an AA 6061 workpiece. This occurred because the depth of cut was mistakenly set higher than

Tool Information		Workpiece	Machine
ID	Description (Model)		
A*	6.35 mm diameter 2-flute square end mill (Kennametal, 2SE0250IX150A)	AA 6061, AISI 1018	*imi_vm20i*, *imi_vmx30ui*
B*	50.8 mm diameter 3-insert shell mill (Insert: Kennametal, ZDET16M504FR721) (Shell mill: Kennametal, C5720VZ16-A2.00Z3R)	AA 6061	*imi_vm20i*, *imi_vmx30ui*
C1	50.8 mm diameter 4-insert shell mill (Insert: Kennametal, HNGJ0604ANERGM) (Shell mill: Kennametal, KSHR200HN4345C3)	AA 6061, AISI 4140	*imi_vm20i*
C2	50.8 mm diameter 5-insert shell mill (Insert: Kennametal, XDPT090412SRGP) (Shell mill: Kennametal, C7792VXD09-A2.00Z5R)	AISI 1018, AISI 4140	*imi_vmx30ui*
D*	12.7 mm diameter 4-flute square end mill (Kennametal, H1TE4RA0500R100HAR030)	AA 6061, AISI 4140	*imi_vm20i*, *imi_vmx30ui*
E*	12.7 mm diameter 5-flute square end mill (Kennametal, UGDE0500J5BRB)	AA 6061, AISI 4140	*imi_vm20i*, *imi_vmx30ui*
F1	12.7 mm diameter 3-flute square end mill (Kennametal, ABDE0500J3AS)	AA 6061	*imi_vm20i*
F2	12.7 mm diameter 5-flute square end mill (Kennametal, KOR5SE0500R150SLC)	AA 6061	*imi_vmx30ui*
G1	63.5 mm diameter 6-insert shell mill (Carbide insert)	ASTM A36	*tmf_vf10*
G2	50.8 mm diameter 8-insert 30° lead angle shell mill (Carbide insert)	ASTM A36	*tmf_vf10*

Table 2. Tool specifications with corresponding workpiece materials and machines in the MSM dataset.

the maximum cutting capacity of Tool F1. As a result, chip adhesion was also observed on Tool F1 after the same operation, as presented in Fig. 9(d). These cases highlight how unintended conditions during machining can propagate from fixture or process instabilities to tool damage and ultimately to workpiece defects. In some experiments, once workpiece slippage leading to tool breakage or chip residuals on workpiece due to improper cutting parameters had already occurred (as in Fig. 9), the operation was terminated early either to protect the machine or because no further cutting was possible. All these failure cases were included in the MSM dataset to ensure comprehensive coverage of cutting anomalies.

Workpieces were typically oriented with the w dimension facing forward (negative y-direction) and h aligned vertically, although alternative orientations were employed depending on tool geometry (diameter, length). Within a single depth pass, spindle speed, cutting direction, feed rate, and a_e were varied, while the same a_p was maintained throughout that pass. Across different passes and workpieces, multiple a_p values were included within the allowable range for each material, thereby ensuring that diverse cutting conditions were represented in a single workpiece. Additional tests were conducted by varying the clamping position of the tool in the holder (long vs. short setup). In both *imi_vm20i* and *imi_vmx30ui*, a dataset generally corresponds to machining a single workpiece without tool changes.

The experimental design intentionally incorporated a wide range of tools, workpiece materials, and machining parameters to enhance the generalizability of the MSM dataset. This diversity captures realistic machining scenarios, supports robust model development, and enables evaluation under unseen conditions, thereby strengthening the dataset's value for both benchmark research and industrial deployment.

For the industrial setting (*tmf_vf10*), data were collected from real production of a consistent part made from ASTM A36 bar stock ($2039.6 \times 139.7 \times 27.9$ mm) on a shop floor. As in the lab, part probing was conducted prior to every operation. Two tools (G1 and G2) were used, following the routing path shown in Fig. 8(c), After completing the first machining pass with Tool G1, the tool was changed to G2 at the starting point, which repeated the same routing path. Both operations were performed in down-milling direction. For Tool G1, which has a square machining cross section, a_p and a_e are defined directly. For Tool G2, with a 30° lead angle, a_p is defined as the maximum depth of engagement, as illustrated in the schematic of Fig. 8(c) bottom. Since these data were collected in an actual manufacturing environment, operator interventions during machining occasionally occurred. In this dataset, such cases correspond to reductions in feed rate through CNC feed override, and the adjusted parameters are also included in the dataset.

Data curation and labeling process. In constructing the MSM dataset, the primary goal was to preserve raw data as much as possible to maximize flexibility for model development and future research. At the same time, data were curated to ensure ease of use, particularly by aligning and synchronizing multiple data sources. This curation ensures that all sensor modalities can be directly compared and jointly analyzed on a common timeline.

In the laboratory setting, data acquisition programs were manually initiated for each experiment, corresponding to one NC program machining one workpiece with a single tool. As shown in Fig. 5(a), MTConnect data (machine controller, power meter, vibration sensors) were stored in CSV format. Each MTConnect record was automatically timestamped at the moment of acquisition. Sound sensor data were recorded as continuous

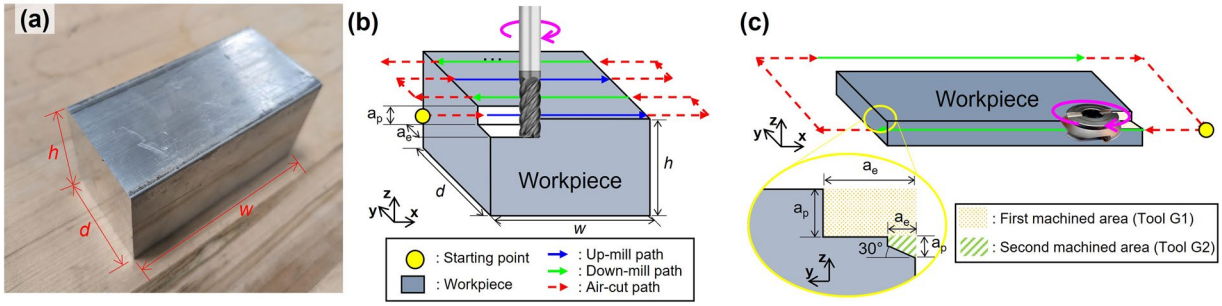


Fig. 8 Workpiece and cutting paths: (**a**) Photograph of the workpiece (AA 6061) with dimensions, (**b**) Schematic of the tool paths and cutting parameters for *imi_vm20i* and *imi_vmx30ui*, and (**c**) Schematic of tool paths and cutting parameters for *tmf_vf10*.

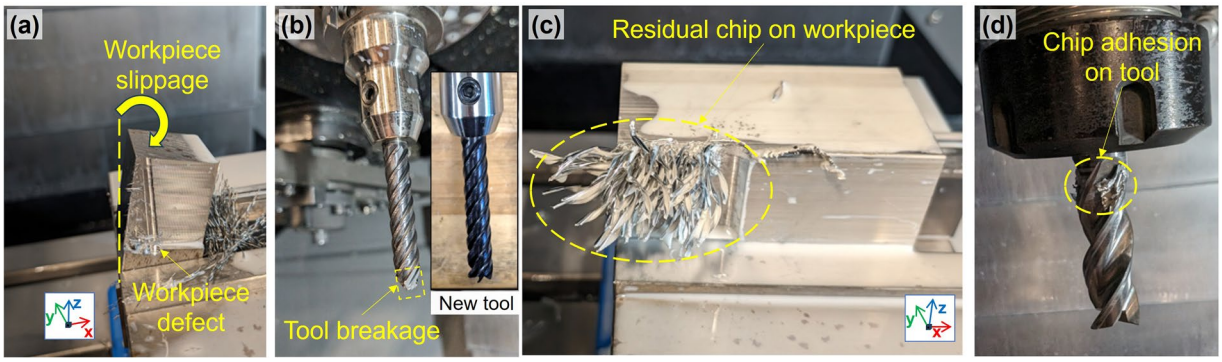


Fig. 9 Examples of cutting anomalies and catastrophic failures: (**a**) Workpiece slippage leading to surface damage, (**b**) Tool breakage of Tool E induced by chatter and slippage from (**a**), (**c**) Residual chip formation on AA 6061 workpiece, and (**d**) Chip adhesion on Tool F1 observed after machining in (**c**).

start	end	label	direction	N	ap	ae	F	Z	D	Tool	Workpiece
10.517	15.765	1	UP	10000	19.05	0.1905	20.32	2	6.35	A*	AA 6061
17.557	22.805	1	DOWN	10000	19.05	0.1905	20.32	2	6.35	A*	AA 6061
26.773	32.021	1	UP	11000	19.05	0.1905	20.32	2	6.35	A*	AA 6061
33.792	39.04	1	DOWN	11000	19.05	0.1905	20.32	2	6.35	A*	AA 6061
…											

Table 3. First few rows of label.csv file from the *imi_vm20i dataset1*.

WAV files, with filenames set to the acquisition start time. Accelerometer signals were also saved in CSV format, with each row containing a measurement value and its timestamp.

To facilitate usability, all datasets were synchronized by aligning timestamps to a common zero point at the start of each experiment. Specifically, one data stream was designated as the reference (set to 0 seconds), and all other streams were cropped accordingly to ensure a common starting point across modalities within each dataset. Therefore, the first column of MTConnect and accelerometer CSV files is time (in seconds), and WAV files can be traced to the same timeline based on the known sampling rate (48 kHz). Based on this synchronized time information, a label.csv file was created for each dataset. An example of the first four rows from *imi_vm20i dataset1* is shown in Table 3. The column definitions and units are as follows:

- start: Start time of cutting path (s).
- end: End time of cutting path (s).
- label: 0 = Normal, 1 = Abnormal, or 2 = Tool defect.
- Note: Even with a defective tool, if cutting itself was anomalous, the label was set to 1. If the tool defect did not cause anomality, the label was set to 2.

- direction: Cutting direction, UP (up-milling) or DOWN (down-milling).
- N: Spindle speed (RPM).
- ap: Depth of cut (mm).
- ae: Width of cut (mm).

- F: Feed rate (mm/s).
- Z: Number of flutes.
- D: Tool diameter (mm).
- Tool: Tool ID (* indicates defective tool).
- Workpiece: Workpiece material.

Cutting start and end times were initially approximated using the NC program together with MTConnect position information. These sources provide a coarse indication of when the tool engages and disengages the workpiece based on the programmed tool path and machine position. The estimates were then manually refined by examining synchronized sensor signals, particularly the sound recordings and other sensor data, to ensure accurate labeling of the actual cutting intervals. Because the sound signals were recorded at 48 kHz, they provided fine temporal cues for identifying the start and end of cutting. This refinement was necessary because variations in workpiece material size could cause slight differences in the actual cutting start and end times. It was particularly important in cases involving failures that led to early termination, ensuring that the start and end times of each cutting path were correctly identified.

Annotation of cutting anomalies by experts (shown in Fig. 1) in the lab datasets was conducted as follows. Two researchers who conducted the experiments collaborated with a machining expert to review sensor signals, machining parameters, and recorded sounds for each cutting path. The annotation criteria were:

- 0 (Normal): Stable cutting without anomalies.
- 1 (Abnormal): Cutting conditions exhibiting unstable or undesirable machining behavior that should be stopped to protect the tool, workpiece, or machine.
    - These conditions may arise from multiple causes, such as excessive tool load, setup errors, fixture instability, tool wear, chatter, or forced vibration, either individually or in combination.
- 2 (Tool defect): Pre-existing tool defects (e.g., broken insert/flute, excessive wear) that were intentionally introduced. Note that label 2 only indicates a defective tool operating under normal cutting conditions, not an anomaly requiring stoppage.

In this dataset, the "Abnormal" label does not denote a single vibration mechanism. Rather, it refers to cutting conditions judged by experts to deviate from stable normal machining and to require intervention, based on the combined review of synchronized sensor signals, machining parameters, and recorded sounds. The label is intended to support data-driven anomaly detection under realistic machining scenarios rather than a strict classification of individual vibration mechanisms. The primary focus of anomaly detection would be label 1 (abnormal cutting). The backgrounds of annotators for the lab datasets (*imi_vm20i* and *imi_vmx30ui*) are summarized below:

- Expert A: 10 years of academic research in CNC machining.
- Expert B: 3 years of academic research in CNC machining.
- Expert C: 17 years of industrial experience as CNC operator, CAM programmer, and toolmaker.

For the industrial dataset (*tmf_vf10*), data extraction was based on the machine's execution state. Operations where the MTConnect *Execution* event was ACTIVE were extracted, defining the start and end times of each part machining cycle. Each operation was treated as a dataset unit. Sensor signals were synchronized to the same 0-second reference: CT data were downloaded as CSV files from AWS InfluxDB, and sound data were available as WAV files from BOX cloud storage, as shown in Fig. 5(b). Start and end times were refined through cross-checks similar to the lab data. Because industrial operations have relatively long continuous cutting paths (approximately 2–4 minutes), segments within these paths were further inspected, and intervals judged to exhibit cutting anomalies were explicitly annotated. Annotation criteria were the same as for the lab datasets. Experts A and B, who had installed sensors and managed data collection at the TMF Center, contributed to the annotation together with two on-site machining experts. Cross-validation among annotators was used to finalize the labels. The industrial experts' backgrounds are:

- Expert D and E: 20+ years of industry experience in CNC machining.

Additionally, for the laboratory datasets, videos were recorded through the machine's automatic door window using a smartphone at the start of each experiment. These videos are provided alongside the dataset to aid understanding and interpretation of the experiments and may also serve as a resource for video-based model development.

Data Records

The dataset is available on Kaggle[30] at https://doi.org/10.34740/kaggle/ds/8392825.

Structure and file types. Figure 10 shows the directory structure and example data files of the MSM dataset. Each dataset folder (e.g., *imi_vm20i*/dataset1) contains synchronized sensor files, the corresponding NC program (for laboratory datasets), and an annotation file ('label.csv'). The available sensor files depend on the machine configuration and may include accelerometer data ('acc.csv'), sound recordings ('s0.wav', 's1.wav', 's2.wav'), MTConnect machine data ('mtc.csv'), and current transformer data ('ct.csv'). For laboratory datasets, an

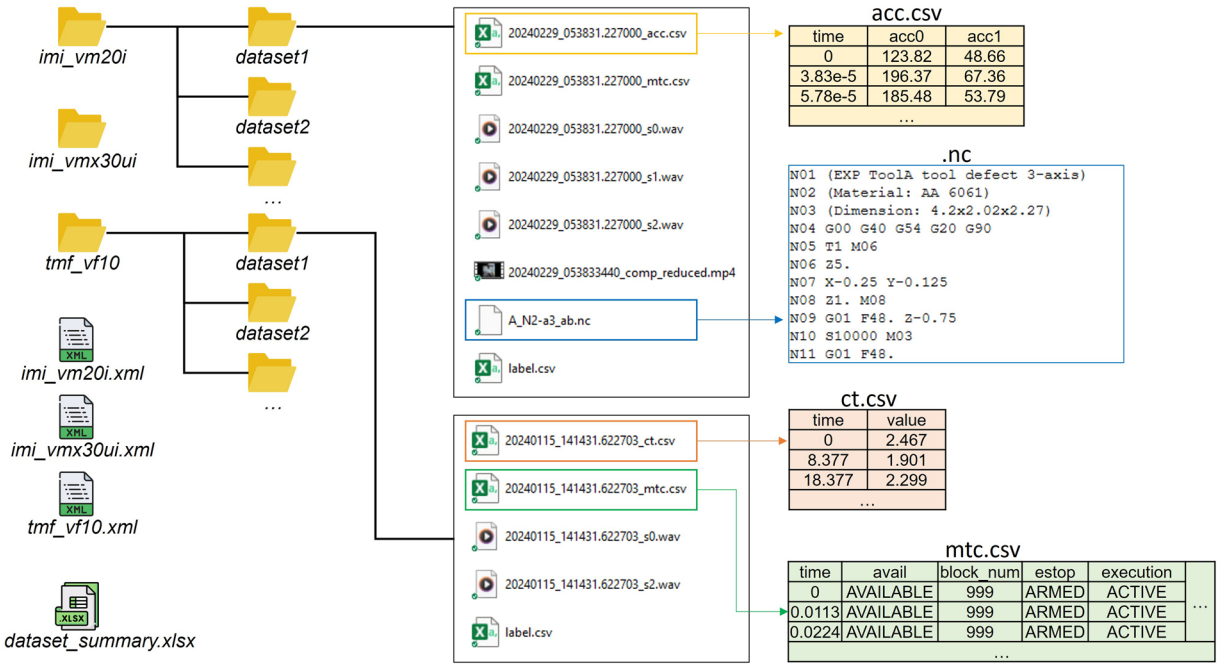


Fig. 10 Directory hierarchy and example data files of the MSM dataset, illustrating machine-level folders, dataset folders, and representative file types (accelerometer, sound, MTConnect, current transformer, NC program, label file, and experiment video).

Machine	Label 0 (Normal)	Label 1 (Abnormal)	Label 2 (Tool defect)
imi_vm20i	69.17	26.33	4.50
imi_vmx30ui	64.48	27.71	7.80
tmf_vf10	97.49	2.51	0.00

Table 4. Distribution of cutting labels (percentage) across machines in the MSM dataset.

experiment video file ('.mp4') recorded during the machining process is also provided. For the industrial dataset (*tmf_vf10*), the NC code is not provided due to production confidentiality.

Filenames of data follow a consistent rule: the reference timestamp used for synchronization (in yyyymmdd_HHMMss.ssssss format), followed by an underscore (_), the data type suffix, and the file extension. The suffix indicates the data modality: 'acc' for accelerometer data, 'mtc' for MTConnect machine data, 'ct' for current transformer data, and 's0', 's1', and 's2' for sound recordings from *Sound0*, *Sound1*, and *Sound2*, respectively. On the right side of Fig. 10, examples of file contents are shown, including 'acc.csv', '.nc', 'ct.csv', and 'mtc.csv'. In all CSV files, the first column is time, recorded in seconds. An example of 'label.csv' is presented in Table 3, and the same format is applied across all datasets.

At the root directory, along with machine directories, the dataset provides 'dataset_summary.xlsx' and MTConnect information models (e.g., 'imi_vm20i.xml'). The 'dataset_summary.xlsx' file summarizes the tool-workpiece combinations used in each machine dataset, allowing users to quickly identify dataset composition. In addition, the Remark column records reasons for early stops during experiments; for the *tmf_vf10* dataset, expert comments on cutting anomalies are also included in the Remark column.

Data volume and summary table. The MSM dataset spans a total of 8 hours 12 minutes 23 seconds, of which 58% (4 h 44 m 38 s) corresponds to cutting time. By machine, the durations are 3 h 16 m 28 s for *imi_vm20i*, 1 h 37 m 19 s for *imi_vmx30ui*, and 3 h 18 m 35 s for *tmf_vf10*. Cutting anomaly label distributions for each machine are summarized in Table 4.

Technical Validation

In the laboratory environment, all computers involved in data acquisition (Fig. 5(a)) were synchronized to the same NIST time server using Network Time Protocol (NTP). This ensured that all recorded data streams shared a common timestamp reference. For MTConnect data integration, MTConnect Standard-compliant information models were designed and included in the dataset, enabling consistent mapping of MTConnect data items and supporting further applications. The integrated MTConnect information models (Device.xml) for each machine were validated for the compliance using the True Analytics Manufacturing Solution® Validator (https://validator.tams.ai). This guarantees that the semantics and vocabulary of the MTConnect data conform

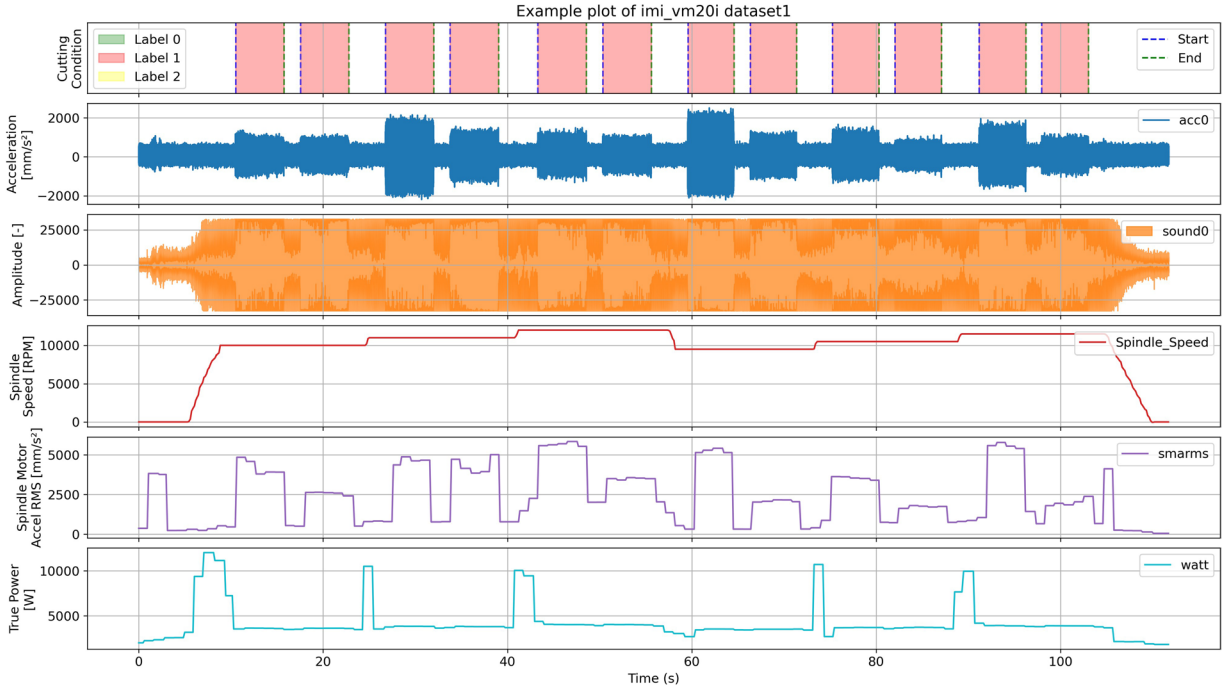


Fig. 11 Example plot of dataset1 from *imi_vm20i* machine tool.

to the standard. All MTConnect and sensor data were further checked during annotation and processing to verify that no missing values or corrupted files were present. For labeling, multiple experts reviewed the same machining signals and discussed each case to determine whether the cutting paths should be labeled as normal or abnormal according to the defined criteria.

As an example, Fig. 11 presents a subset of raw data from dataset1 of the *imi_vm20i*. The plot on the top shows the annotated cutting intervals, where the start and end times from 'label.csv' (Table 3) are marked and the corresponding cutting anomaly label are highlighted. The second plot shows the accelerometer signal (*Acc0*) mounted on the spindle, followed by the microphone signal (*Sound0*) located inside the machine enclosure. Since plotting raw waveforms of sound signals at the full sampling rate results in visual saturation due to the large number of samples, we employed an envelope-based downsampling method (min/max per window), a common practice in audio signal visualization. The remaining plots visualize selected MTConnect data streams: spindle speed recorded from the machine controller, spindle motor acceleration RMS (*smarms*), and true power consumption (*watt*) from the power meter. This representation preserves the perceptual dynamics of the signal while enabling clear alignment with cutting labels. No additional unit conversion was applied. The key point of this visualization is that heterogeneous sensor data sources are well aligned along a common time axis, enabling direct comparison and multi-modal analysis.

Data availability

The dataset described in this Data Descriptor is publicly available on Kaggle[30] at https://doi.org/10.34740/kaggle/ds/8392825 and is openly accessible under the Creative Commons Attribution 4.0 International (CC BY 4.0) license.

Code availability

No custom code was used to generate or process the data described in this Data Descriptor. All data collection and synchronization were performed using standard acquisition systems prior to publication, and no proprietary or custom-written code was employed.

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Competing interests

The authors declare no competing interests.

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